

LOCAL-FIRST · EXPLAINABLE · EXPORTABLE

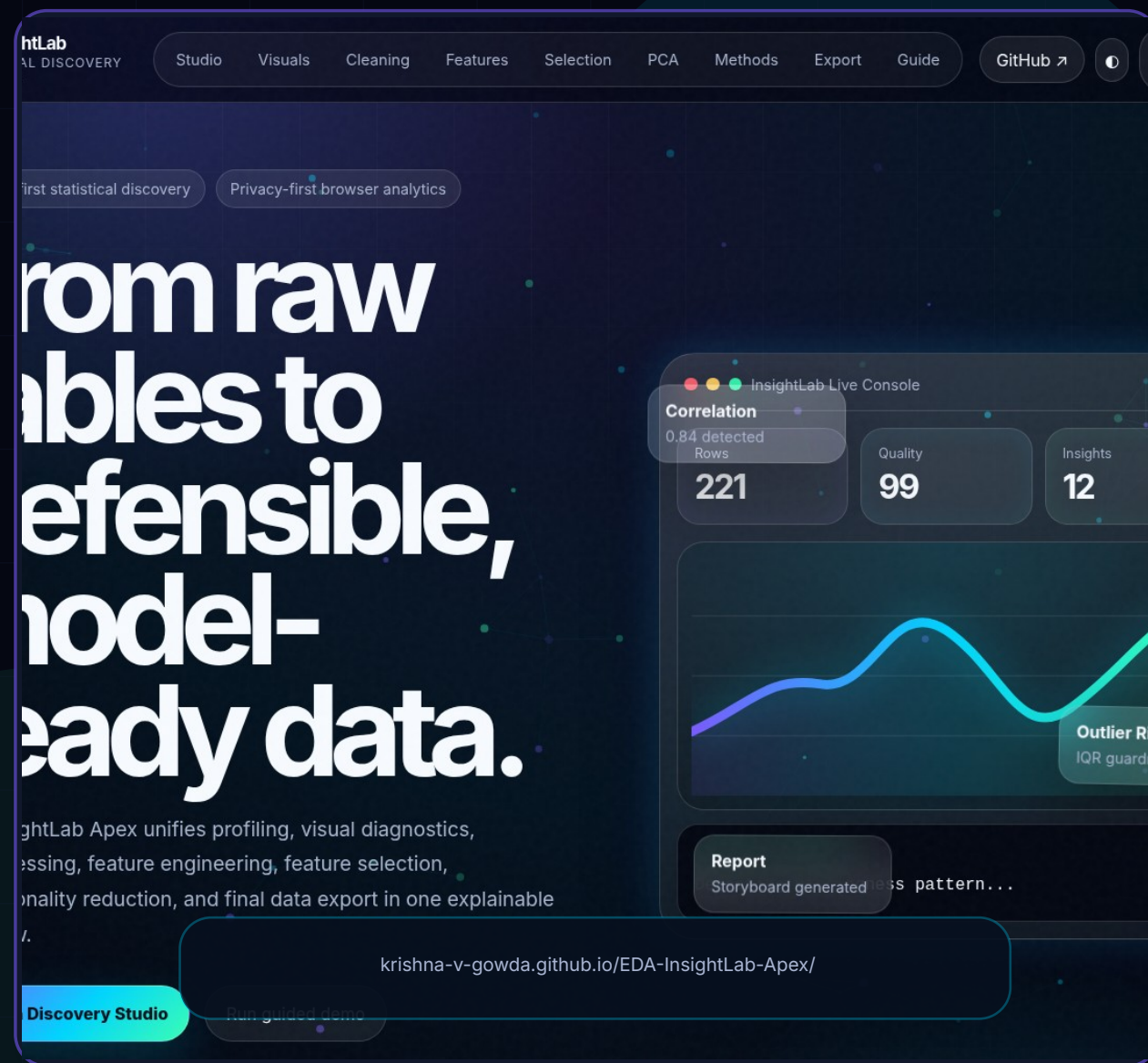
EDA InsightLab Apex

From raw tables to defensible,
model-ready data.

A statistical discovery product for profiling, visual diagnostics, preprocessing, feature engineering, target-aware selection, PCA, and final analytical export.

KRISS GOWDA

AI · DATA SCIENCE · PRODUCT ENGINEERING



The model is rarely the first failure point.

Most analytical risk enters earlier - through data quality, leakage, weak feature design, and opaque handoffs.

01

MESSY INPUT

Missing values, duplicate records, inconsistent types and extreme values distort the evidence.

02

TARGET LEAKAGE

Future or target-derived information can make a model look strong while making it useless in reality.

03

WEAK SIGNALS

Raw columns often fail to express the domain relationships a model actually needs.

04

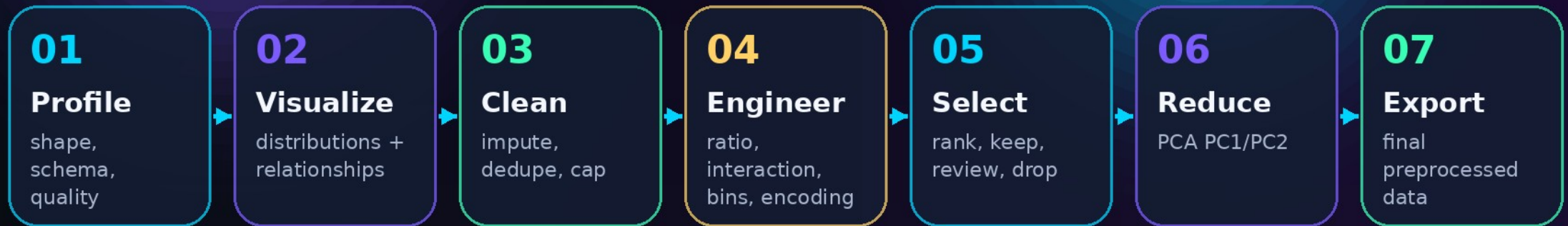
OPAQUE HANDOFF

When transformations live across ad hoc scripts, stakeholders cannot trace what changed or why.



A better product must protect the entire path - not only display the final chart.

One visible journey from raw CSV to analytical handoff.



Showcase story: not just EDA charts - a complete, transparent, model-ready pipeline.

Every transformation is explainable and exportable, so the final data can move into modeling, reporting or decision-making.

A command center for understanding the data before changing it.

CSV

Drop dataset here

Local-only processing. No upload to any server.

Sample dataset

Manufacturing Quality Process

Load Dataset

Refresh

Search preview

Try: urban, series A, premium

Focus column

Temperature_C

X axis

Temperature_C

Y axis

Pressure_bar

Chart family

Histogram

Dataset Health Score

99

/100

- Missingness is 0.5% across 2,873 cells.
- 1 duplicate row found.
- Highest missing column: Humidity_pct (4.5%).
- Strongest outlier pressure: Yield_pct (0.9%).
- Highest correlation: Defect_Rate_pct ↔ Yield_pct (-0.68).

Copy executive summary

Insight Engine

Strong data foundation

The current dataset health score is 99/100, which is suitable for a public demo and early analysis.

Missingness pattern

Humidity_pct has the highest missingness at 4.5%. Explain whether the absence is random or linked to collection process.

Relationship signal detected

Defect_Rate_pct and Yield_pct show a strong negative correlation (-0.68). This is a strong

DATA PREVIEW

First 24 records

221 matching rows

Run_ID	Production_Line	Shift	Operator_ID	Material_Batch	Temperature_C	Pressure_bar	Line_Speed
MQ001	Line D	Morning	O-31	Batch M3	70.18	28.68	400
MQ002	Line B	Morning	O-31	Batch M4	68.41	30.91	420
MQ003	Line C	Night	O-31	Batch M4	77.44	31.73	479
MQ004	Line A	Evening	O-17	Batch M4	75.09	30.74	399

searchable preview

1 Local-first intake

Built-in datasets or a user CSV. Rows are processed inside the browser.

2 Transparent health signals

Missingness, duplicates and IQR pressure are visible - not hidden inside a pipeline.

3 Narrative layer

The interface explains what the diagnostics mean and what should be reviewed next.

Heuristic score ≠ model accuracy

Charts that teach, not just decorate.



Context

Distribution
Histogram + summary statistics

Spread
Box plot + IQR diagnostics

Relationship
Scatter + Pearson correlation

Structure
Correlation heatmap

Data quality
Missingness map

Association is evidence - not causation.

Transformations become a product workflow – not invisible notebook state.

The interface shows a workflow for feature engineering. On the left, under 'Numeric feature B', 'Pressure_bar' is selected. Under 'Categorical feature', 'Run_ID' is selected. Below these are buttons for 'Create Ratio Feature', 'Create Interaction Feature', 'Create Polynomial Feature', 'Bin Numeric Feature', 'One-Hot Encode Category', and 'Add Z-Score Scaled Features'. The middle section shows two steps: '2 Feature engineering' (Created interaction feature Temperature_C_x_Pressure_bar) and '3 Feature engineering' (Created polynomial feature Temperature_C_squared). Below these are buttons for 'Download final CSV', 'Pipeline JSON', and 'Python pipeline'. The right section shows the resulting features: 'Temperature_C^2', 'Temperature_C_x_Pressure_bar' (Interaction feature), and 'Temperature_C_per_Pressure_bar' (Ratio feature).

Ratio

A / B with zero-safe handling

Interaction

A × B for combined effects

Polynomial

A² for curvature

Binning

Quantile-based segments

Encoding

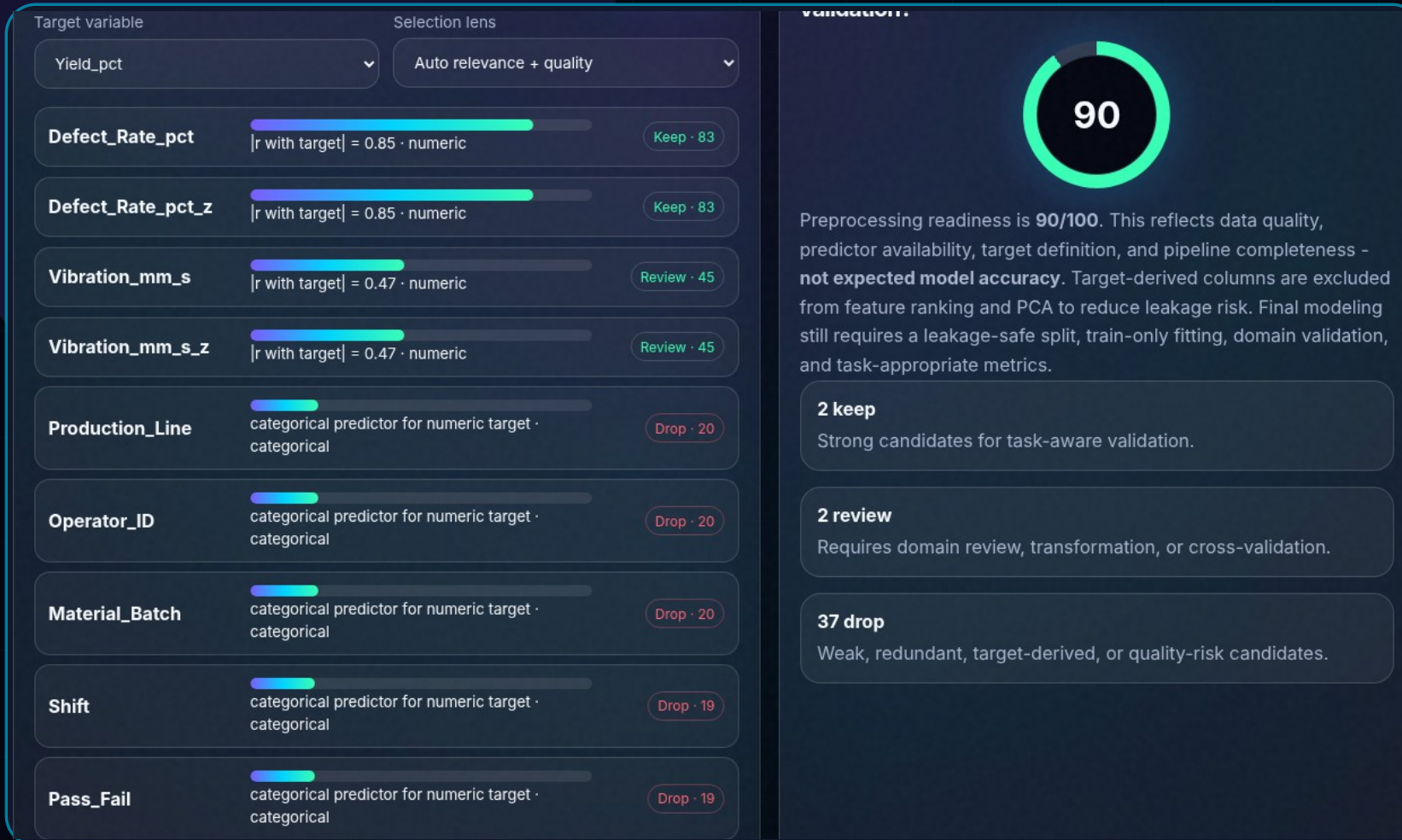
Low-cardinality predictor dummies

Scaling

Standardized predictor copies

Target protected during predictor transforms

Target-aware screening with visible leakage guards.



1 Meaningful default target

Manufacturing opens with Yield_pct, not an arbitrary final column.

2 Association by data type

Pearson, eta-squared, and Cramer's V support mixed tabular data.

3 Leakage protection

Target and target-derived columns are removed from predictor ranking and PCA.

Screening heuristic → domain review → cross-validation

Compress correlated numeric structure without hiding the method.

PRINCIPAL COMPONENT ANALYSIS

2D multivariate map

Run PCA

Add PC columns

PCA projection of standardized predictor features



PCA used 10 numeric predictor features after excluding the selected target and target-derived columns. PC1 explains 36.1% and PC2 explains 24%, giving a compact multivariate map for clusters, leverage points, and dimension reduction.

QUALITY + DOE LENS

quality lens

Process and experiment diagnostics

Response / quality metric

Yield_pct

Lower spec limit

50

Upper spec limit

90

Calculate capability + DOE surface

Capability for Yield_pct

Mean = 84.27, Std = 2.3, Cp = 2.89, Cpk = 0.83.

DOE-style response thinking: Defect_Rate_pct and Yield_pct show $r = -0.85$. In a design-of-experiments discussion, these can be treated as candidate factors to investigate response movement, not proof of causality.

Interpretation: This module links EDA to quality engineering by moving from distribution to specification limits, process capability, and factor-response reasoning.

e components.

1

STANDARDIZE

Comparable feature scales

2

COVARIANCE

Capture joint variation

3

EIGENVECTORS

Approximate principal directions

4

PROJECT

Map rows into PC1/PC2

PCA supports compression and exploration - it does not identify causal drivers.

The analytical handoff includes data, code, metadata, and narrative.

Processed data table preview (220 rows · 45 columns):

	Production_Line	Shift	Operator_ID	Material_Batch	Temperature
1	Line D	Morning	O-31	Batch M3	70.18
2	Line B	Morning	O-31	Batch M4	68.41
3	Line C	Night	O-31	Batch M4	77.44
4	Line A	Evening	O-17	Batch M4	75.09
5	Line C	Morning	O-17	Batch M2	71.9
6	Line D	Morning	O-58	Batch M1	69.17
7	Line D	Night	O-22	Batch M2	72.3
8	Line C	Evening	O-22	Batch M1	75.1

Buttons: Download processed CSV, Download discovery report

Statistical Discovery Assistant: Explain the analysis

Ask: Which features should I keep? Is my data model-ready? What changed after preprocessing?

Try asking: "What changed after preprocessing?" or "Which feature selection result should I present?"

or review:

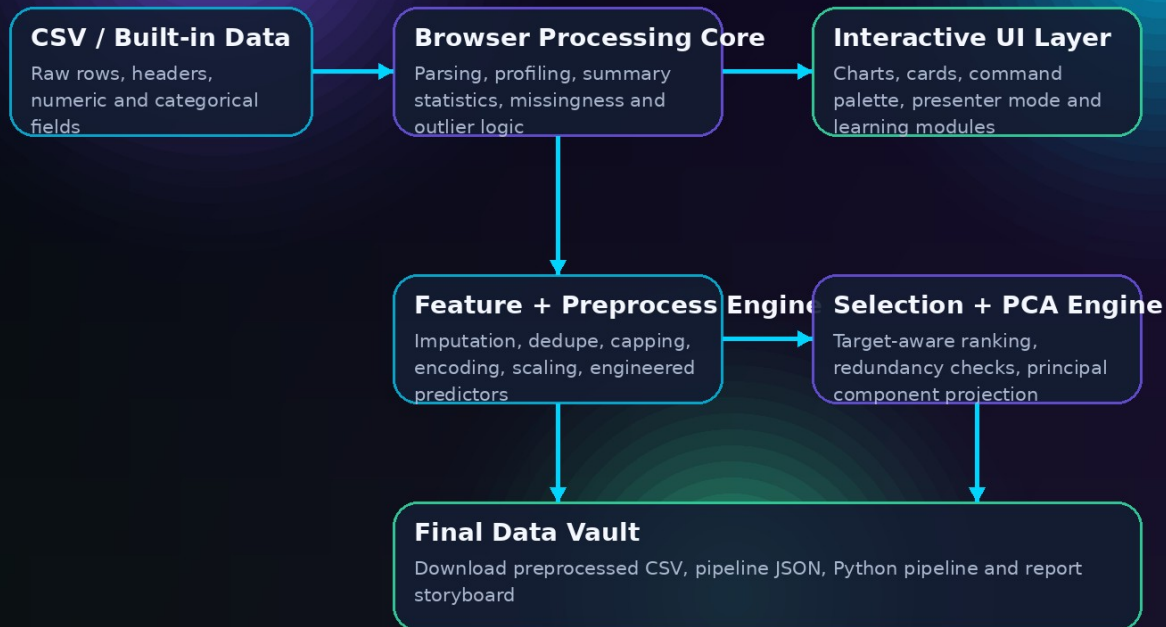
- CSV**: Processed rows and engineered features
- JSON**: Ordered transformation metadata
- PYTHON**: Leakage-aware pipeline template
- REPORT**: Discovery summary and method caveats

Processed data is a strong analytical starting point - final modeling still requires a leakage-safe split, train-only fitting, and task-specific validation.

Zero-backend by design. Production-ready evolution by roadmap.

EDA InsightLab Apex - System Architecture

Offline-first statistical discovery pipeline for learning, preprocessing and final data export



CURRENT RELEASE

- Static HTML, CSS, and JavaScript
- Browser-memory data state
- SVG chart rendering
- Client-side file exports
- No project-owned data API

PRODUCTION EVOLUTION

- Validated Python statistics service
- Larger datasets and job queues
- Authentication and data lineage
- Versioned runs and audit logs
- Governed AI explanations

Trust comes from separating what works from what still needs validation.

The product is strongest when its claims are precise, testable, and honest.

VERIFIED WORKFLOW

- CSV ingestion and schema inference
- Descriptive profiling and SVG visuals
- Tracked cleaning and engineered features
- Target protection and leakage guards
- Feature screening and PCA workflow
- CSV, JSON, Python, and report exports
- Responsive desktop and mobile layout

VALIDATION BOUNDARIES

- Health and readiness scores are heuristics
- Selection is screening, not trained importance
- IQR flags are diagnostics, not deletion rules
- Correlation is not causation
- PCA needs reference-library comparison
- No production model is trained in v1.0
- Regulated decisions require stronger controls

EVIDENCE OVER HYPE

From flagship prototype to governed analytical platform.

A clear roadmap converts the current product into a serious foundation for AI, operations, quality, and decision intelligence.

NOW

v1.0

Local-first discovery, feature workflow, PCA, export

NEXT

v1.1

Reference-library tests, schema overrides, accessibility

THEN

v1.2

Leakage-safe splits, baseline models, error analysis

SCALE

v2.0

Python service, versioning, auth, lineage, audit

INTELLIGENCE

v2.5

Governed AI copilot grounded in computed evidence

BUILD WITH EVIDENCE. SHIP WITH INTENT.

Kriss Gowda · github.com/Krishna-V-Gowda · krishna-v-gowda.github.io/EDA-InsightLab-Apex/

Questions · feedback · collaboration